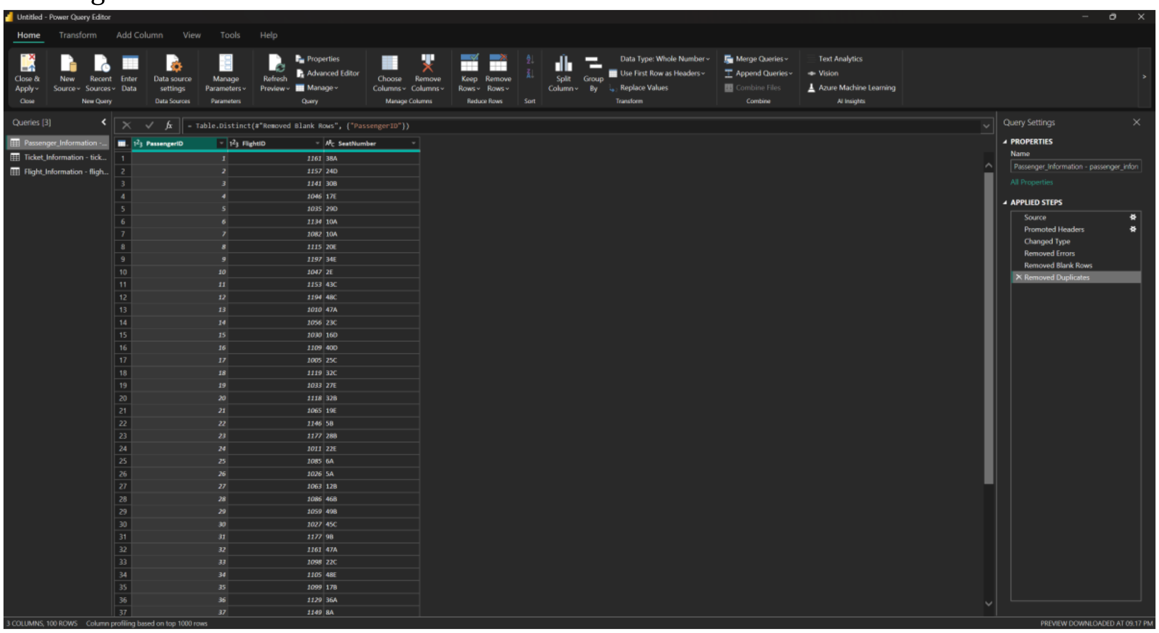
Airline Data Management and Analysis Using Power BI

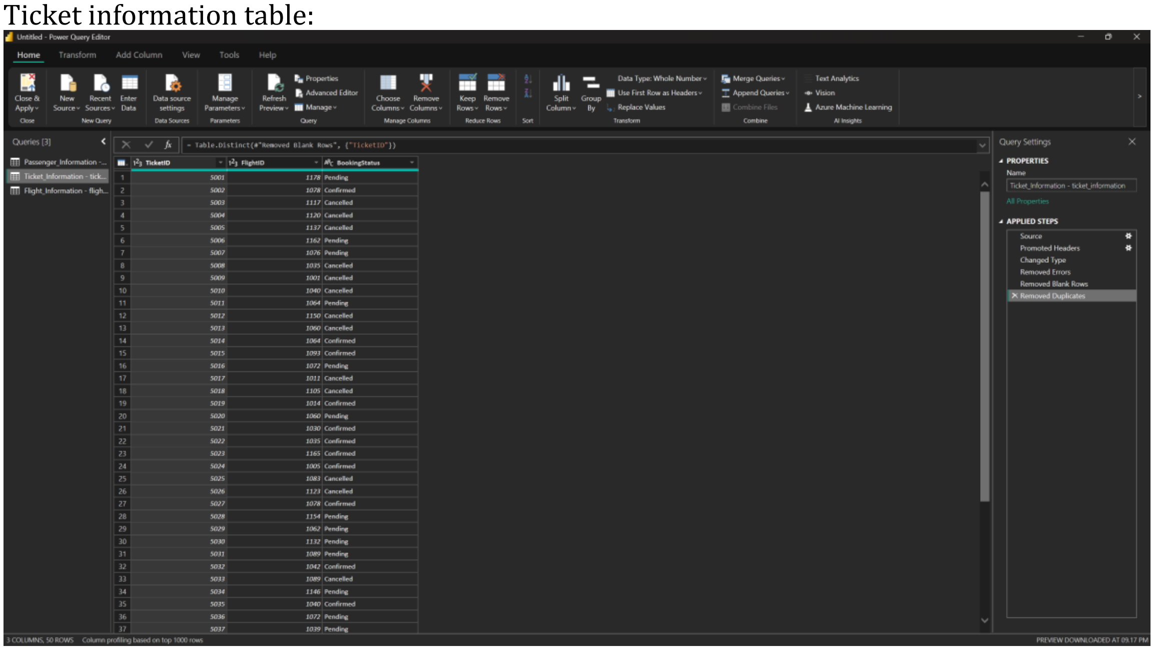
Task 1 : Data Preparation and Cleaning

I imported the Flight Information, Passenger Information, and Ticket Information datasets into Power Query Editor.
I removed duplicate entries, handled missing values, and formatted columns properly to ensure clean and usable data for analysis.

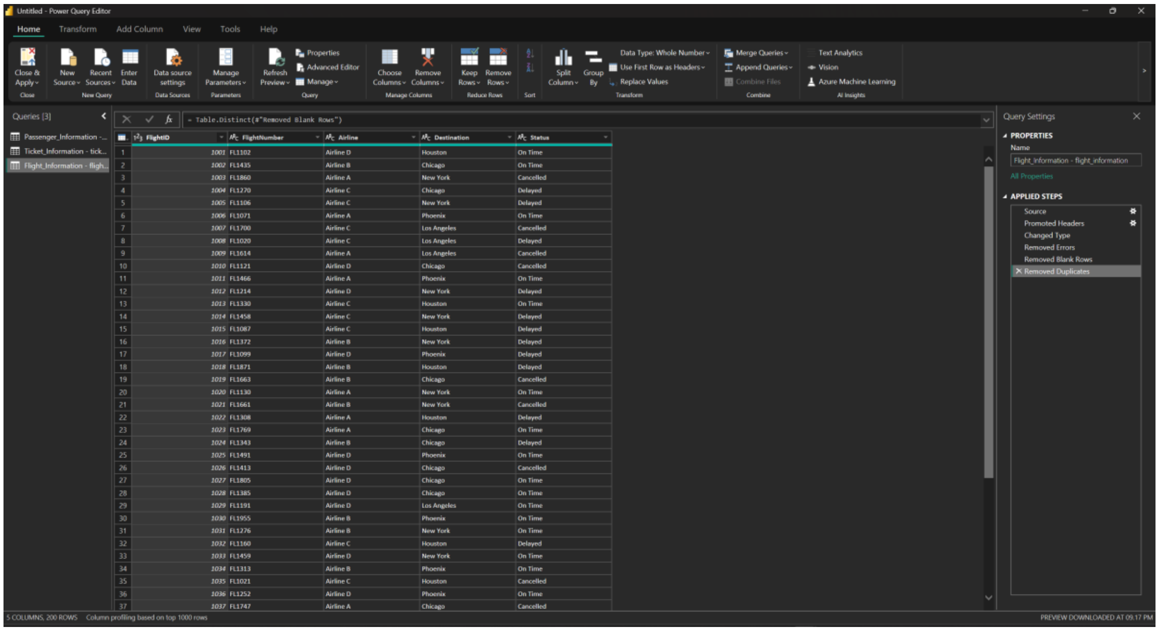
Passenger information table:



Ticket information table:



Flight information table:

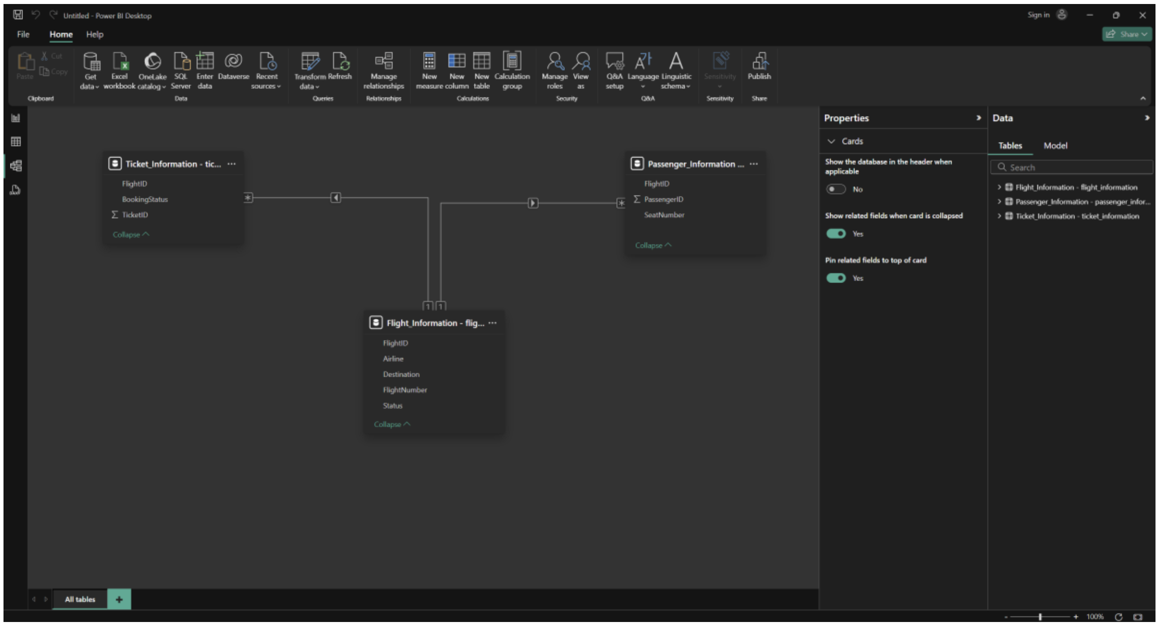


Insight: Cleaning the data improved its reliability and ensured that all further analysis would be accurate and error-free.

Task 2: Data Modeling

I created relationships between the datasets using FlightID as the primary key.
I configured the correct cardinality and ensured proper cross-filtering between the tables to create a robust data model.

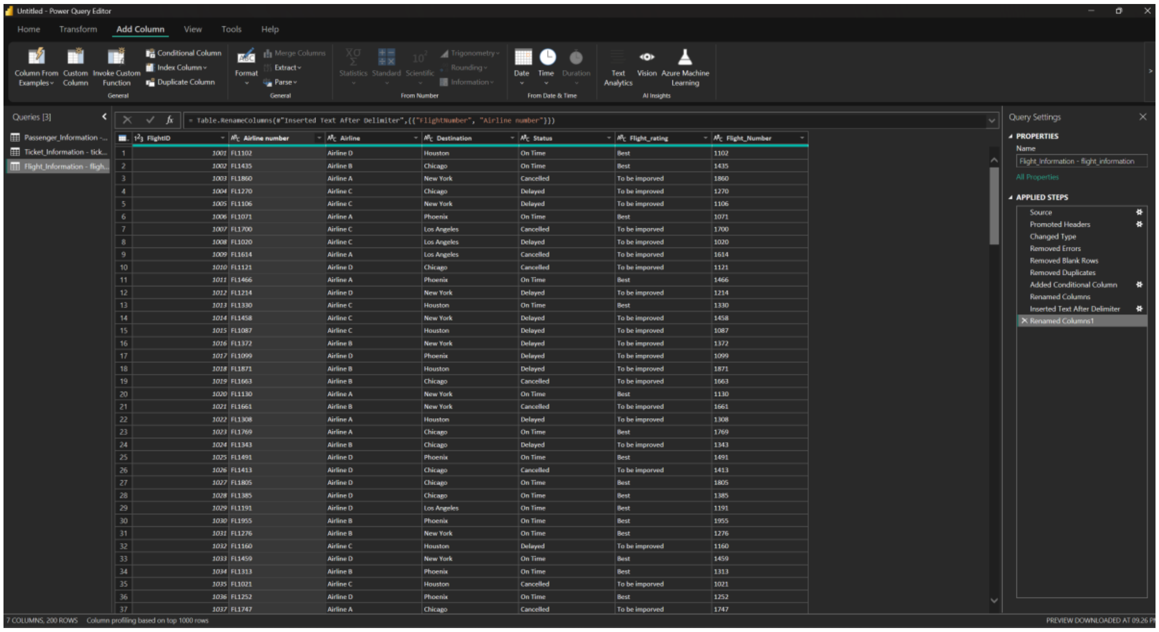
Relationship model view:



Insight: Establishing accurate relationships made it easy to connect flight, passenger, and ticket information for meaningful cross-analysis.

Task 3: Enhanced Data Insights

I added a conditional column classifying flights as "Best" or "To Be Improved" based on their status.
I also used the "Column from Examples" feature to extract numeric parts of the FlightNumber for cleaner visualizations and easier data manipulation.



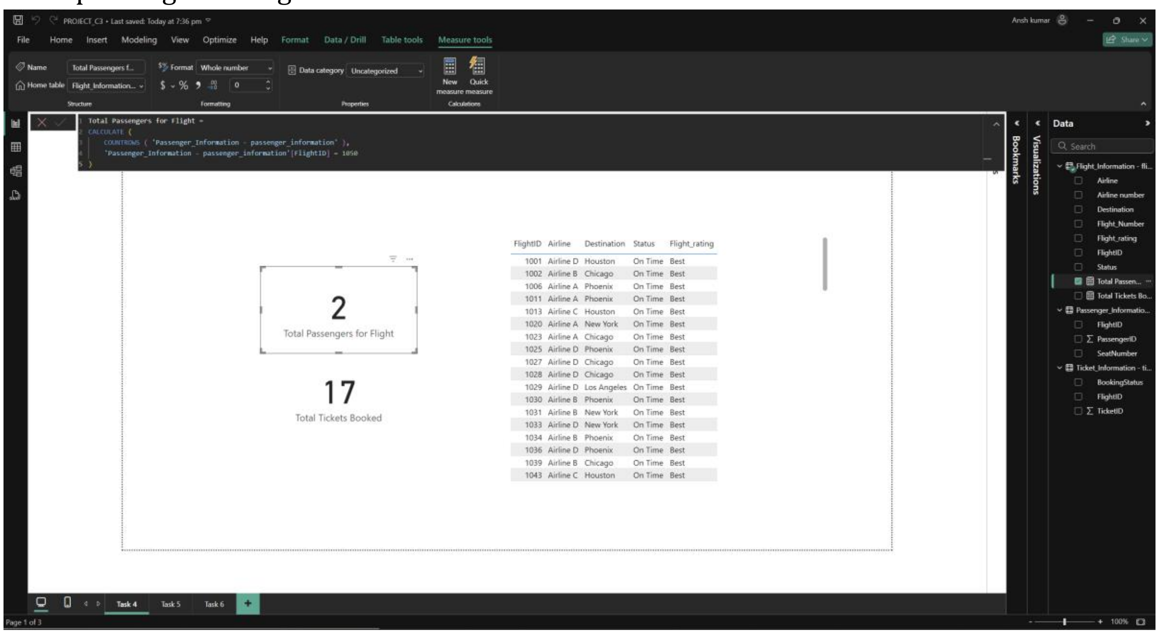
Insight: Creating new columns helped classify and organize the flights, making trend analysis much more straightforward.

Task 4: Calculations Using DAX

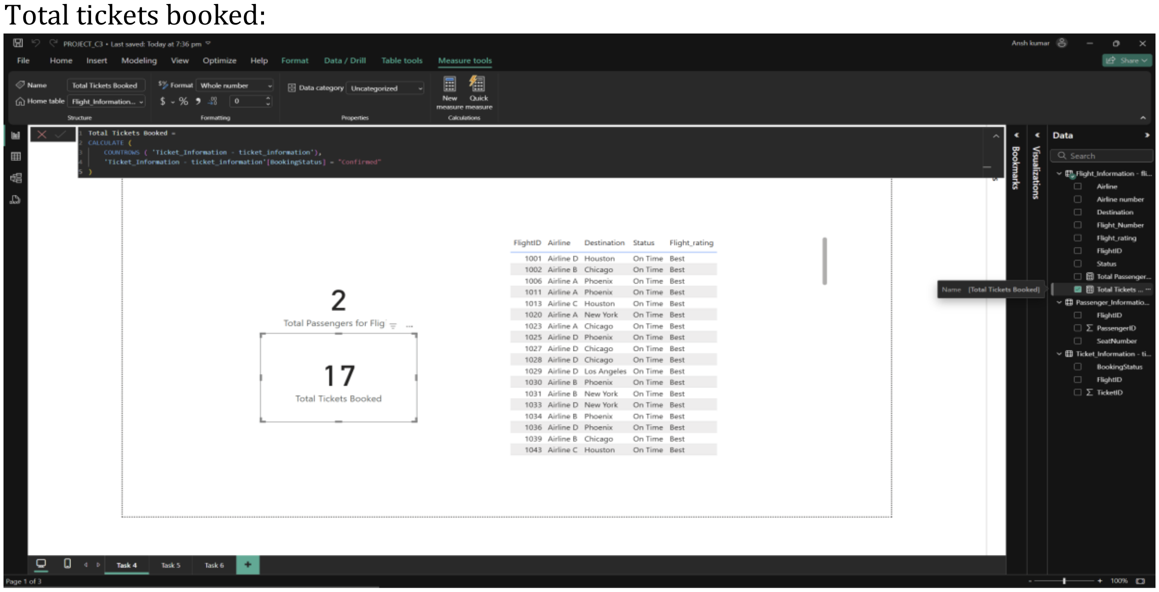
I wrote DAX measures to calculate:
- Total passengers per flight
- Total tickets booked
- A filtered table showing only the "Best" flights

These calculations allowed for deeper exploration into operational efficiency and booking trends.

Total passenger for flights:



Total tickets booked:

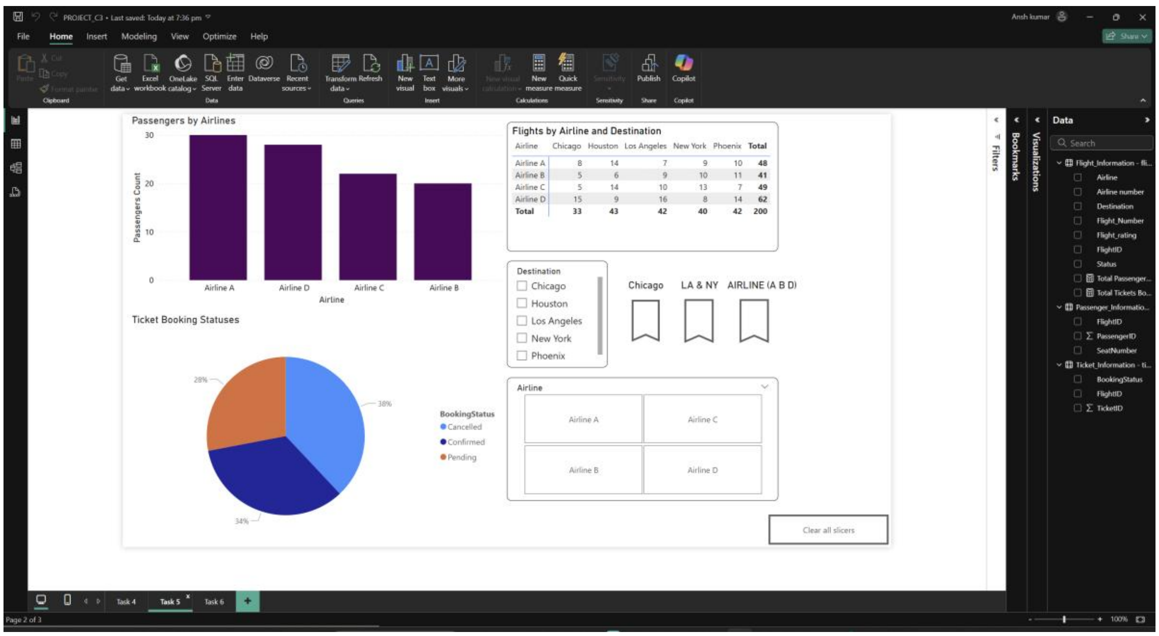


Insight: Using DAX unlocked dynamic calculations that provided sharper insights into passenger load and flight performance.

Task 5: Visualization and Interactive Features

I created various visuals, including:
- Passenger count by airline
- Ticket booking status overview
- Flights categorized by airline and destination

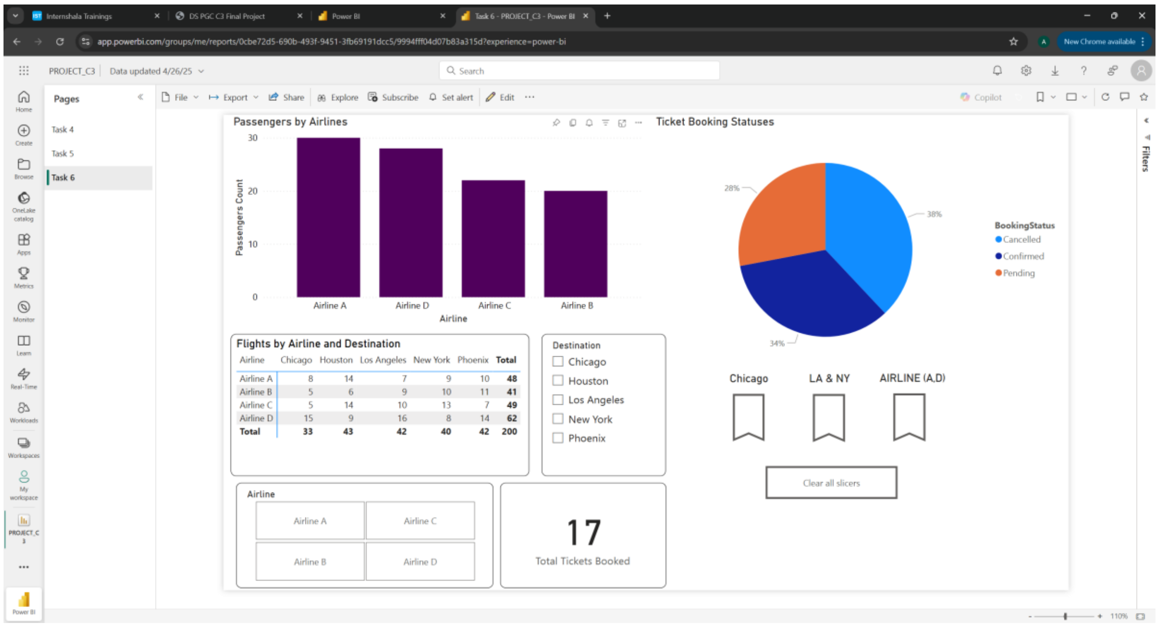
Interactive features such as slicers for Destination and Airline were added, along with dedicated airline-specific features for deeper individual analysis.



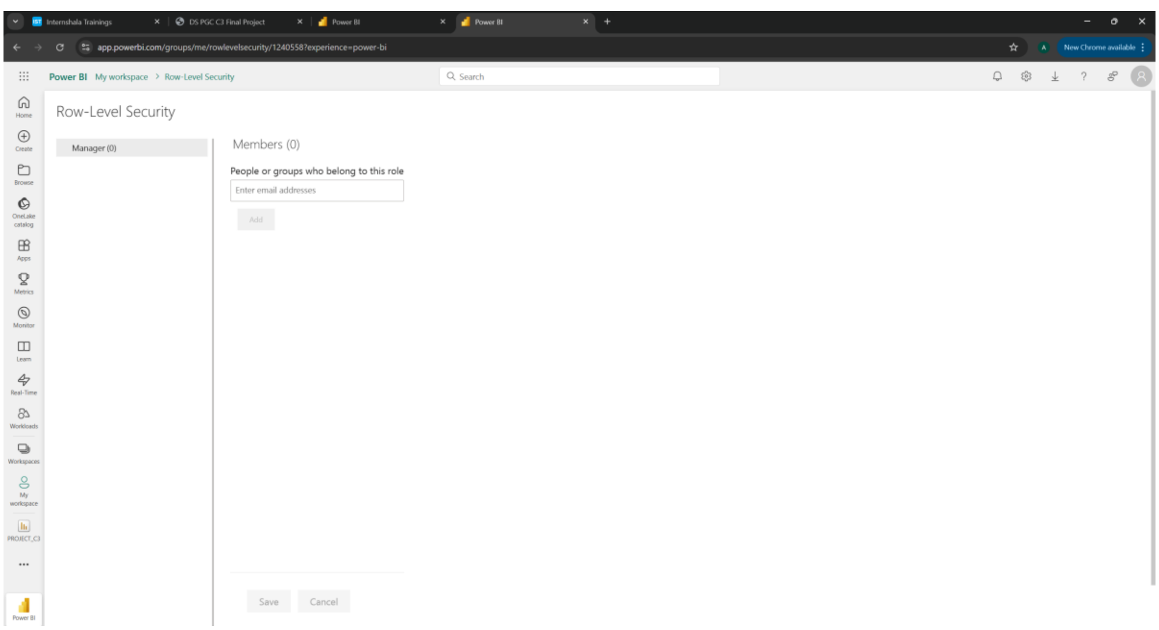
Insight: Visualizing the data revealed hidden trends, and adding interactivity made it easier for users to explore specific details on their own.

Task 6: Final Dashboard and Power BI Service

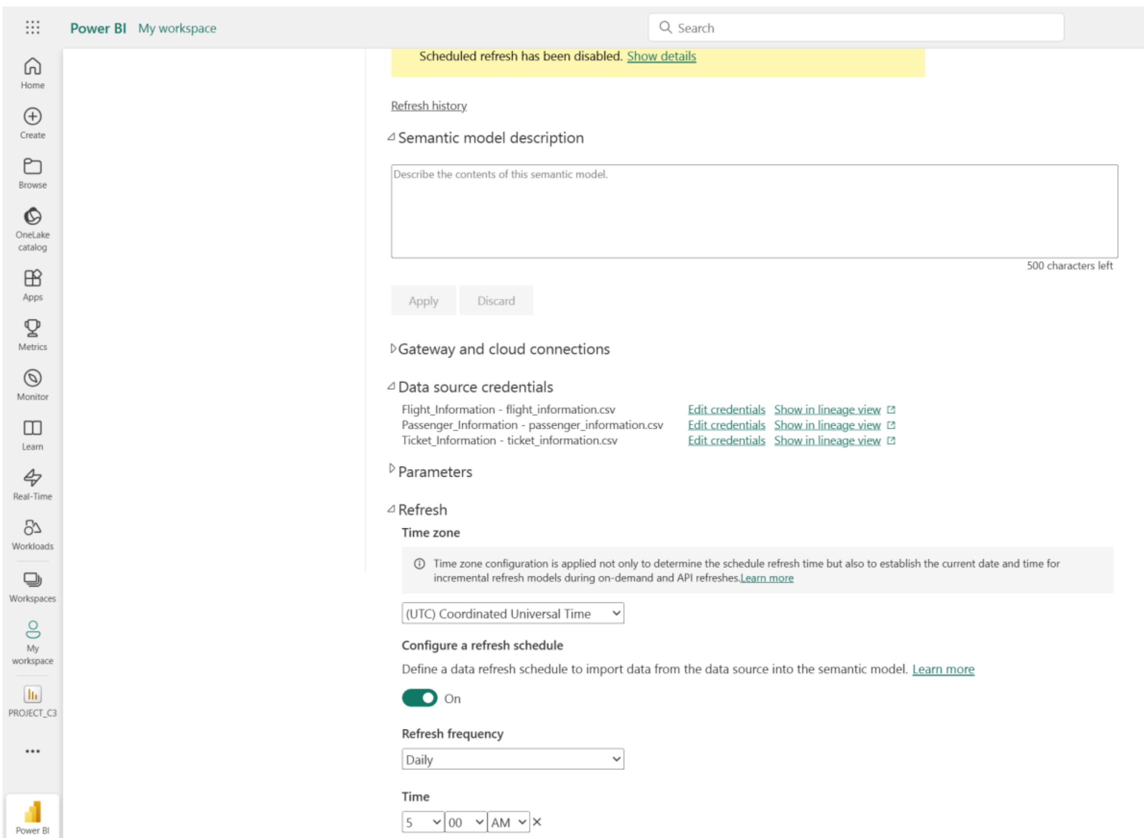
I designed a final comprehensive dashboard that compiled all major visuals and insights cohesively.
I also configured Row-Level Security (RLS) for Airline A and scheduled a daily refresh at 5 PM to keep the dashboard data current.



Enabled RLS



Scheduled refresh



Insight: Publishing the dashboard with security and automatic refresh enhanced its professionalism and made it ready for real-world enterprise use.

[End of Report]
[Ansh kumar]

Video Explanation:

https://drive.google.com/drive/folders/1RAXYOuvcT-ATD2552MjY-2vJxYbo3dLP?usp=drive_link